10. Marginal Revenue: $MR = \frac{dR}{dx}$

11. Profit: $P(x) = R(x) - C(x)$

12. Elasticity: $\eta = \frac{x}{y} \cdot \frac{dy}{dx}$

13. Elasticity of demand: $\eta_d = -\frac{p}{x} \cdot \frac{dx}{dp}$

14. Elasticity of supply: $\eta_s = \frac{p}{x} \cdot \frac{dx}{dp}$

15. Relationship between MR, AR and η_d :

$$MR = AR\left[1 - \frac{1}{\eta_d}\right] \text{ (or) } \eta_d = \frac{AR}{AR - MR}$$

16. Marginal function of y with respect to x (or) Instantaneous rate of change of y with respect to x is $\frac{dy}{dx}$

17. Average cost $[AC]$ is minimum when $MC = AC$

18. Total revenue $[TR]$ is maximum when $MR = 0$

19. Profit $[P(x)]$ is maximum when $MR = MC$

20. In price elasticity of a function,

    (a) If $|\eta| > 1$, then the function is elastic

    (b) If $|\eta| = 1$, then the function is unit elastic

    (c) If $|\eta| < 1$, then the function is inelastic.

21. $EOQ = q_0 = Rt_0 = \sqrt{\frac{2C_3R}{C_1}}$

22. Optimum number of orders per year

$$n_0 = \frac{\text{demand}}{\text{EOQ}} = R\sqrt{\frac{C_1}{2C_3R}} = \sqrt{\frac{RC_1}{2C_3}} = \frac{1}{t_0}$$

23. Minimum inventory cost per unit time, $C_0 = \sqrt{2C_1C_3R}$

24. Carrying cost = $\frac{q_0}{2} \times C_1$ and

    ordering cost = $\frac{R}{q_0} \times C_3$

25. At EOQ, Ordering cost = Carrying cost

26. If $u(x,y)$is a continuous function of x and y then, $\frac{\partial^2 u}{\partial y \partial x} = \frac{\partial^2 u}{\partial x \partial y}$

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